

Early detection of Chronic Kidney Disease Using Machine Learning

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Submitted by

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1. INTRODUCTION

1.1 Project Overview

In recent times a lot of people have been suffering from a lot of various serious ailments because of their busy lifestyle. One of the most common silent killers is heart attack, which is a result of unhealthy lifestyle.

Another very common chronic disease which may occur due to unhealthy lifestyle is kidney disease. Kidney troubles occur because of overusing painkiller, eating processed food, not drinking enough water etc. Kidney disease may also occur because of diabetes also.

Diabetes is caused because of the building up of sugar in the bloodstream. Sugar builds up in the bloodstream because of the lack of insulin to break down the sugar or because the body has become insulin resistant. Diabetes leads to kidney disease by making fatty acids to store in kidneys and thus leading to the malfunction of kidneys which will eventually lead to the kidney failure.

But this can be prevented if the kidney disease is diagnosed at an early stage which will not lead to kidney failure and there won't be any need for transplantation of kidney.

The main objective of this project is to create a model which will detect kidney at an early stage and stop the patient from having any further implications.

1.2 Purpose

The main objective of this project is to detect kidney disease at an early stage and start treatment for it as soon as possible so that the patient does not face any further implications.

The model is designed in such a way that it detects kidney disease at an early stage to save the patient from kidney failure.

The model for prediction of the kidney disease at an early stage is built using machine learning algorithms.

2. Literature survey

A literature review is a type of academic writing that provides an overview of existing knowledge in a particular field of research. A good literature review summarizes, analyses, evaluates and synthesizes the relevant literature within a particular field of research.

2.1 Existing problems:

Presently kidney disease is detected at late stages in many countries leading to loss of precious lives. There are very few means to identify them at an early stage. Most of the user details remain unverified and it's difficult to track the fake users. The user interface of the application is not user friendly and the user must have a device with an android operating system with an active internet connection to interact with this application.

2.2 References:

S.NO	AUTHOR	TITLE	NAME OF THE JOURNAL	VOLUME/ YEAR/ ISSUE	ALGORITHM /METHOD
1.	Abrar, Tahmid Tasnim, Samiha Hossain, Md.Mehrab	Early detection of chronic kidney disease using machine learning	Brac University Journal	Volume 8, Issue 1	Support vector regressor, logistic regressor

2.	Fuzhe Ma, Tao Sun, Lingyun Liu, HongyuJing	Detection and diagnosis of chronic kidney disease using deeplearning- based heterogeneous modified artificial neural network.	Elsevier journal	Volume 111	Support Vector Machine, Multi layer perceptron, back propagation n algorithm
3.	Marwa Almasoud, Tomas E Ward	Detection of Chronic KidneyDisease Using MachineLearni ng Algorithms with Least Number of Predictors	International journal of soft computing	Volume 30, Issue 2	Random forest, Gradient boosting, logistic regression.

The above listed are the existing problems and the solutions given by different members. The main disadvantages are also listed to get into different kind of solutions.

- <https://www.sciencedirect.com/science/article/abs/pii/S0167739X20308128>
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8774382/>
- <https://pubmed.ncbi.nlm.nih.gov/18812692/>

2.3 Problem Statement Definition:

Chronic kidney disease (CKD) is one of the most critical health problems due to its increasing prevalence. Many people are not aware of that disease. Chronic kidney disease, also known as chronic renal disease or CKD, is a condition characterized by a gradual loss of kidney function over time. It includes conditions that damage your kidneys and decrease their ability to keep you healthy by filtering wastes from your blood. Diabetes and high blood pressure, or hypertension, are responsible for two-thirds of chronic kidney disease cases. Anyone can get chronic kidney disease at any age. However, some people are more likely than others to develop kidney disease.

Most people may not have any severe symptoms until their kidney disease is advanced. So they are not able to find the disease at the earlier stage. Hence the severity of the problem increases. A better testing method which could possibly detect CKD in the early stages would be much more useful. Medical test results taken for other purposes are used to detect CKD at early stages. Various efforts have been undertaken to advance early therapy to prevent the condition from progressing to chronic disease.

Recent research suggests that some of the negative outcomes can be avoided with early identification and treatment. Peculiar and contributing attributes from the above-mentioned test results are combined to develop a Machine Learning Model.

This Machine Learning Model will be used to predict CKDs rather early than the presently existing methods.

3. Ideation and proposed solution

3.1 Empathy map canvas

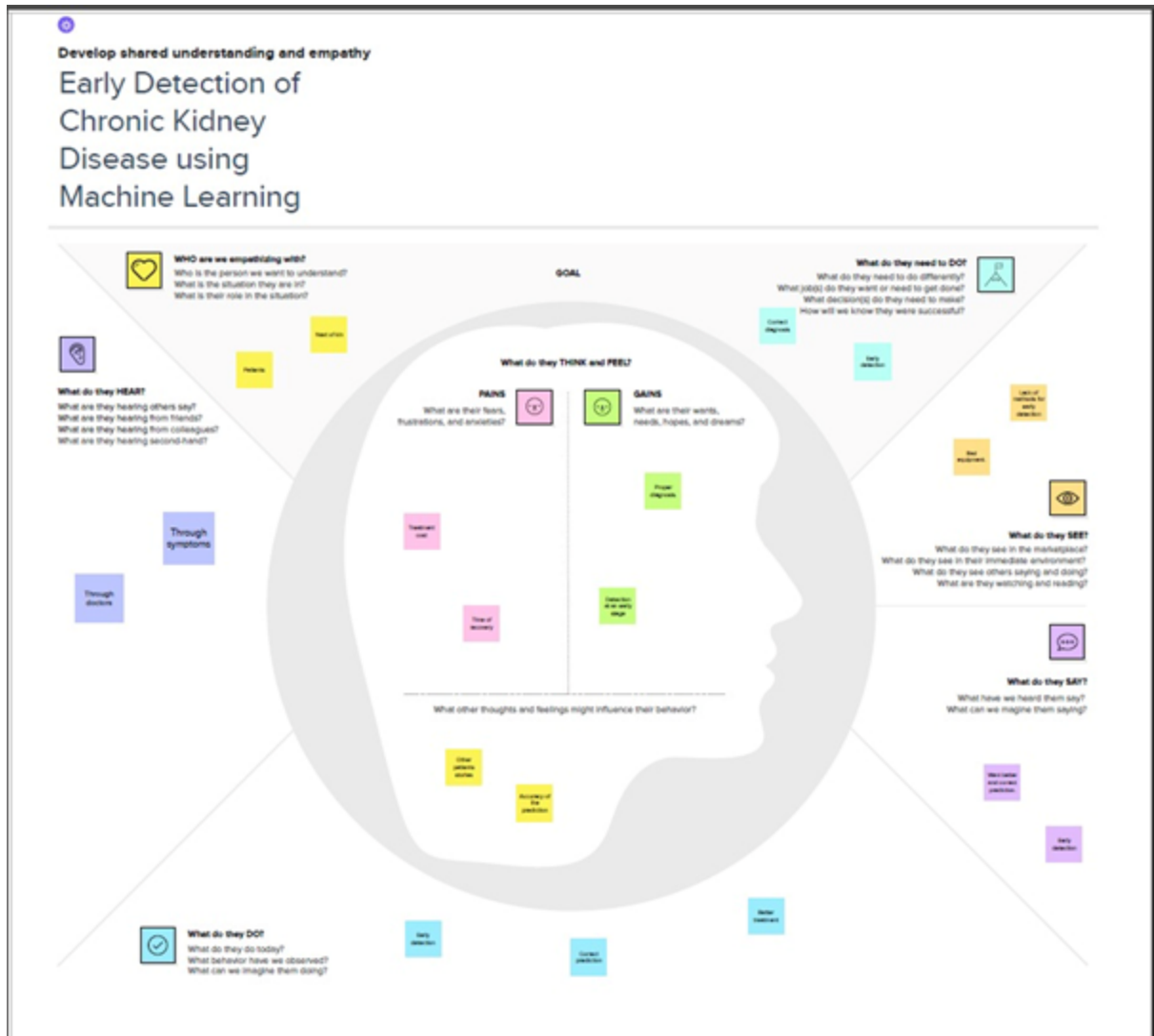


Fig 3.1.1 Empathy map

An empathy map can depict a group of users, such as a consumer segment, in a manner similar to user personas. Teams can utilize an empathy map as a collaborative tool to learn more about their clients.

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviors and attitudes. It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the

user's perspective along with his or her goals and challenges.

For our project we have created an empathy map in which the target group are patients, people who know others suffering from the same ailment, doctors etc. We have asked them for their input by asking them various questions for situations and recorded them.

We then built an empathy map to really understand what other people might be going through because of this ailment and how effectively we can build a model to help them.

3.2 Ideation and Brainstorming

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Ideation is often closely related to the practice of brainstorming, a specific technique that is utilized to generate new ideas. A principal difference between ideation and brainstorming is that ideation is commonly more thought of as being an individual pursuit, while brainstorming is almost always a group activity.

Ideation is the process where you generate ideas and solutions through sessions such as Sketching, Prototyping, Brainstorming, Brain writing, Worst Possible Idea, and a wealth of other ideation techniques.

1

Define your problem statement

This is a textbox...

🕒 5 minutes

PROBLEM

Early Detection of Chronic Kidney Disease
Using Machine Learning

Fig 3.2.1 Define problem statement

The first step is to gather as a group and discuss what is the problem statement. After finalizing the problem statement by brainstorming methods we define the problem statement.

2

Brainstorm

This is a textbook...

10 minutes



3

Group ideas

20 minutes

**Fig 3.2.2 Brainstorm and group ideas**

We sit as a group and brainstorm the various ideas. Each group member will give four unique ideas about how to collect the dataset, what sort of algorithm needs to be selected, what should the end result be like and how should we display it.

After each team member comes up with a unique idea we group the ideas which are alike or are somewhat similar to each other. This is done so that similar ideas can be somehow integrated together and incorporated with each other.

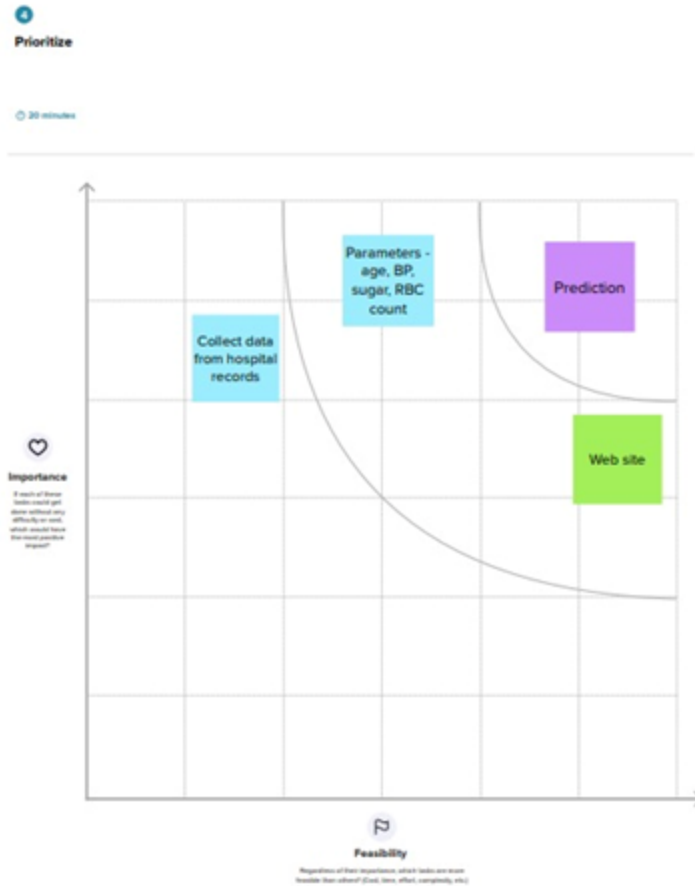


Fig 3.2.2 Idea prioritization

This is the final step in brainstorming and idea generation. After gathering all the unique data the ideas are prioritized. That is some things like the algorithm that is being used is more important than the means of gathering the information.

So we prioritize all the ideas and then make a graph to show which idea has higher priority and which idea has low priority.

3.3 Proposed Solution

Proposed solution means the technical solution to be provided by the implementation agency in response to the requirements and the objectives of the project. It means the combination of software, hardware, other products or equipment, and any and all services (including any installation, implementation, training, maintenance and support services) necessary to implement the solution described by Vendor in its Proposal.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	To predict the presence of chronic kidney disease early in an easier way with the help of known medical history of patients.
2.	Idea / Solution description	Chronic Kidney Disease (CKD) is a major medical problem and can be cured if treated in the early stages. Usually, people are not aware that medical tests we take for different purposes could contain valuable information concerning kidney diseases. Consequently, attributes of various medical tests are investigated to distinguish which attributes may contain helpful information about the disease. The information says that it helps us to measure the severity of the problem and we make use of such information to build a

		machine learning model that predicts Chronic Kidney Disease
3.	Novelty / Uniqueness	• Easier prediction of chronic kidney disease • Well designed model to give the use with the accurate resale value.
4.	Social Impact / Customer Satisfaction	• Patients get to know about the presence of disease early • This model is vital for society as there are a lot of people who are not aware of the presence of the disease.
5.	Business Model (Revenue Model)	Allows patients to know about the disease early which can be cured easily.
6.	Scalability of the Solution	This to make the efforts of knowing the presence of disease which will be helpful for easy curable of that disease.

Fig 3.3.1 Proposed solution

Our problem statement is detection of chronic kidney disease at an early stage with help of any machine learning algorithm. So we have proposed a solution for the problem.

First we define the problem statement in a clear and precise manner. Next we give a brief description of the idea and solution description. After giving a brief description of the idea and solution we mention the novelty of this project. We mention how unique this project is and how there is not any existing system like this.

After finishing the novelty of the project we mention about the social impact the project will have on the society as well as the customer satisfaction.

Then we mention about the business model for the project and finally we discuss the scalability of the solution.

This is what the proposed solution consists of.

3.4 Problem Solution Fit

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It is important to have a good picture of your customer, not only the demographics but preferably also socio graphic data.

The more informed the customer is about their problem, the better. It may even be the case that your customer has realized an inefficient homegrown solution to try to solve the problem. Try as hard as you can to identify how well the customer is aware of the problem and when the customer will run into the problem.

Project Title: Early Detection of Chronic Kidney Disease		Project Design Phase-I - Solution Fit Template		Team ID: PNT2022TMID21290	
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Hospital management, Patients, Doctors, Nurses, Data scientists	6. CUSTOMER CONSTRAINTS CC Large data is to be collected in order to give accurate results in the prediction.	5. AVAILABLE SOLUTIONS AS Various medical tests are available whose analysis leads to finding the chronic kidney disease. Questionnaire about the patient's earlier diagnostics give insight to the early detection.	Explore AS, differentiate	
Focus on J&P, map into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS J&P To predict whether the patient have chronic kidney disease or not, in more accurate and faster way based on certain diagnostic measurements like Blood Pressure (Bp), Albumin(AI).	9. PROBLEM ROOT CAUSE RC One in 7 people are unaware they have chronic kidney disease. It can be cured if treated in the early stages.	7. BEHAVIOUR BE Customer submits the required information, and the algorithm predicts the result.	Focus on J&P, map into BE, understand RC	
Identify Strong TR and EM	3. TRIGGERS TR 10% of the population worldwide is affected by chronic kidney disease, and millions die each year because they do not have access to affordable treatment.	10. YOUR SOLUTION SL An ML Model	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE Search for a suitable dataset → download the dataset 8.2 OFFLINE Preprocess the data → build an ML model → train the model → test the model → Early Detection of Chronic Kidney Disease	Identify Strong TR and EM	
	4. EMOTIONS: BEFORE / AFTER EM Customers can no longer feel stress and can take corrective actions if they are diagnosed early.				

Fig 3.4.1 Problem solution fit

The Problem-Solution Fit is an important step towards the Product-Market Fit, but often an underestimated one. There are ten categories for which we have collected the data to create a problem solution fit for our project.

For each category we sat as a team and came up with multiple solutions which may be useful for our model and then finally selected one solution which will be the most appropriate one for our model.

4. Requirement analysis

4.1 Functional Requirements:

Functional requirements are the desired operations of a program, or system as defined in software development and systems engineering. The systems in systems engineering can be either software electronic hardware or combination software-driven electronics. Functional requirements are product features that developers must implement to enable the users to achieve their goals. They define the basic system behavior under specific conditions.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story/ Sub-Task)
FR-1	Home Page	● Chronic Kidney disease description ● Information about Test Vitals required for prediction ● If new User , REGISTER ● If Already exist, SIGN IN
FR-2	User Registration	● Enters Mail ID and other personal details required for Registering.
FR-3	User Login	● Uses Mail ID and Password for login
FR-4	Test VitalsForm	● Test Vitals should be entered for prediction
FR-5	Result	● If Positive – Test Result along with the Information about what is to be done next will be displayed. ● If Negative – Test result along with preventive measures to prevent themselves from getting Chronic Kidneydisease will be displayed.

4.2 Non Functional Requirements:

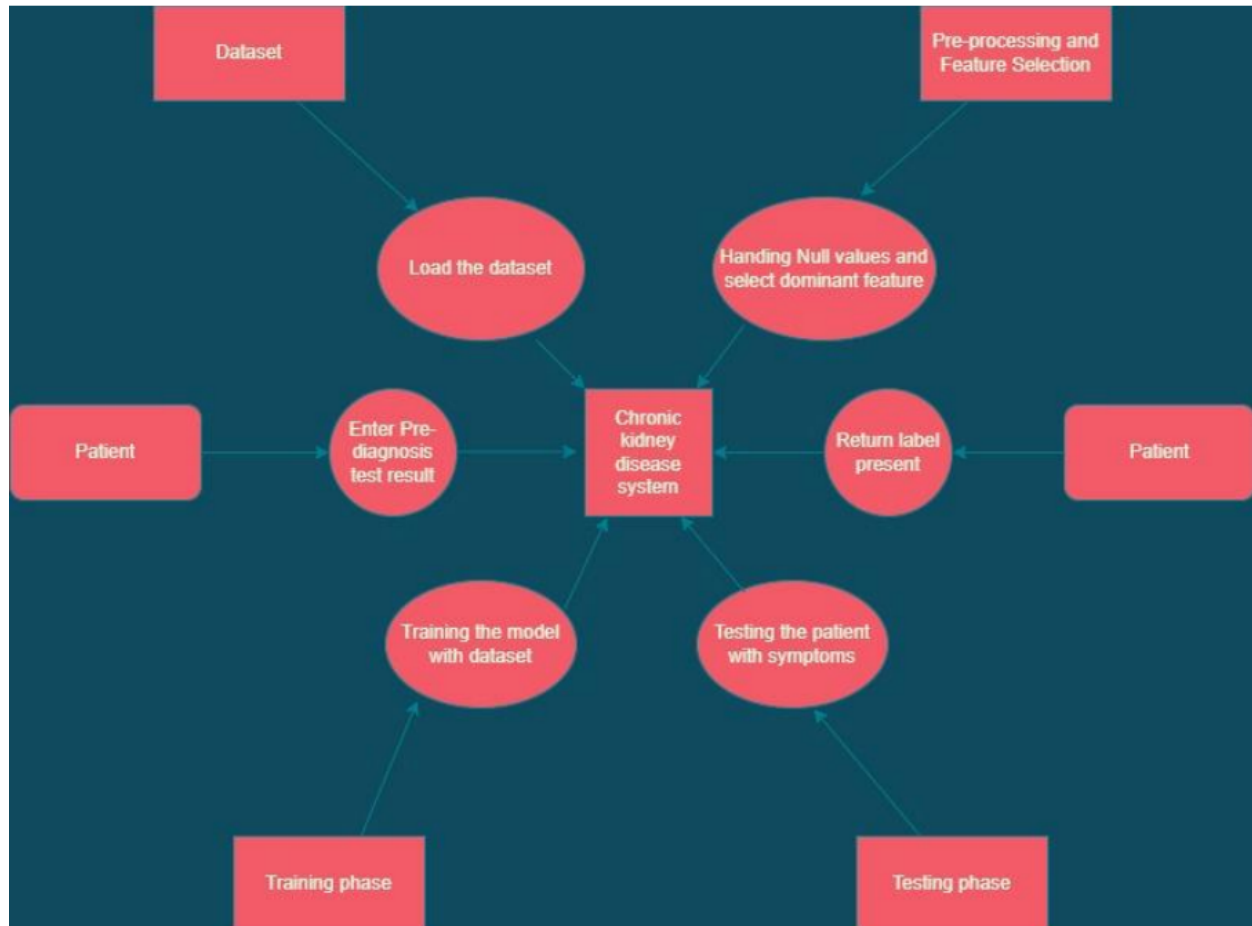
Nonfunctional Requirements (NFRs) define system attributes such as security, reliability, performance, maintainability, scalability, and usability. They serve as constraints or restrictions on the design of the system across the different backlogs. The upcoming list is the non functional requirements for our project.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Even Illiterates and people with no understanding of computer/mobile should be able to use the product.
NFR-2	Security	Access permission for particular system information may be changed by systems data administration.
NFR-3	Reliability	The database update process must roll back all related updates when any updates fails.
NFR-4	Performance	The Home-page load time must be no more than 2 seconds for users
NFR-5	Availability	New Model Deployment must not impact Homepage, test page and result page availability
NFR-6	Scalability	The website Traffic limit must be scalable enough to support 2000,000 users at a time.

5. Project Design

5.1 Data flow diagram

Data flow diagrams are used to represent the flow of data in a business information system graphically. DFD describes the processes that are involved in a system to transfer data from the input to the file storage and reports generation.



Data flow diagrams can be divided into logical and physical. The logical data flow diagram describes flow of data through a system to perform certain functionality of a business. The physical data flow diagram describes the implementation of the logical data flow.

In our project, our first step is to collect the dataset, load it and clean it. After that we will get diagnosis from the patients and run the diagnosis through the model which has been trained and tested for prediction.

5.2 Solution and Technical architecture

Solution architecture provides the ground for software development projects by tailoring IT solutions to specific business needs and defining their functional requirements and stages of implementation. It is comprised of many sub processes that draw guidance from various enterprise architecture viewpoints.

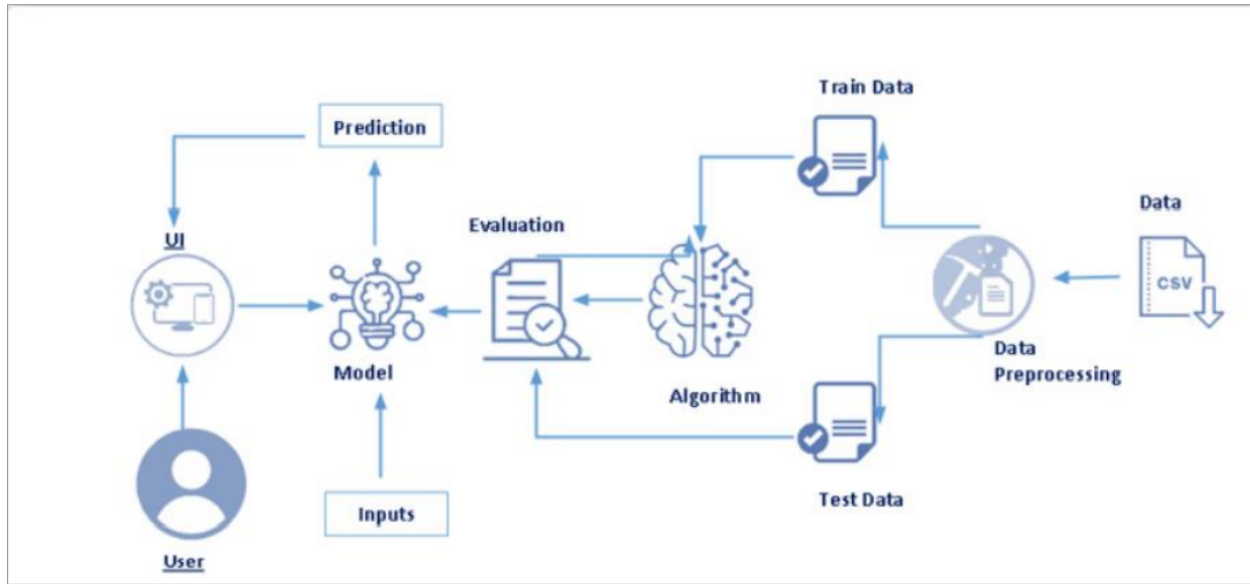


Fig 5.2.1 Technology architecture

The client needs are broadened to include business needs that are somehow connected to technology in solution architecture. These requirements typically become clear after reevaluating current systems to see whether they ultimately help or hurt the firm. Business analysts will occasionally conduct these analyses and define the issue. The next stage is for solution architects to take this issue and begin developing a description of remedies that effectively meet this need.

Solution architecture establishes guidelines and instructions for proper implementation and delivery while translating technical business needs into workable IT solutions. It also takes into account every outside aspect that can affect the growth process. In this manner, there is agreement between EA teams and development teams, and digital initiatives are less likely to fail. When creating and implementing new systems and applications, SA can be thought of as a support system that offers structure and lowers the level of complexity.

5.3 User stories

In software development and product management, a user story is an informal, natural language description of features of a software system. They are written from the perspective of an end user or user of a system, and may be recorded on index cards, Post-it notes, or digitally in project management software.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Webuser)	Registration	USN-1	As a user, I can register for the diagnosis tool using my email and password	I can access my account / dashboard	High	Sprint-1
		USN-2	As a user, I can register for the application through my Gmail	I can register and access the dashboard with my Gmail Login	Low	Sprint-4
	Login	USN-3	As a user, I can log into the application by entering my credentials	I can login and access records	High	Sprint-1
	Dashboard	USN-4	As a user, I can see my records and activities	I can access the functionalities diagnosing tool	High	Sprint-3
	Entry form	USN-5	As a user, I must enter my pre-diagnostic test results	I can use the form to input test results	High	Sprint-2
	Report	USN-6	As a user, I can view the report generated by the tool	I can view negative/ positive results produced after diagnosis	High	Sprint-3
	Remedies	USN-7	As a user, I will receive remedies to treat my symptoms	I can cure my symptoms with the remedies suggested	Medium	Sprint-3
Administrator	Feature importance	USN-8	As an administrator, I should identify the most significant factors that lead to CKD based on the present trend	I must identify important features	High	Sprint-2
	Train model	USN-9	As an administrator, I must use the most suitable ML model for detection of CKD	I should efficiently train the ML model	High	Sprint-2

6. Project planning and scheduling

6.1 Sprint planning and estimation

A strong Agile development team is built on sprints. And the more ready you are before a sprint, the more probable it is that you will achieve your objectives. Planning for spring can assist to prioritize efforts, reduce unexpected events, and (hopefully) ensure that better code is deployed. The sprint, the stage where ideas transform into innovation and valued goods come to life, is the major event in the agile approach. Agile sprints have the potential to be effective and collaborative. They can also be disorganized and ineffective if there is poor planning and direction.

Making a sprint plan is one of the most crucial things you can do to guarantee the success of your efforts because of this.

A sprint estimation shows how much effort a series of tasks require. It's based on assumptions, requirements, and dependencies of a project.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application using email and password.	7	High	Team Lead Member 1 Member 2 Member 3
Sprint-1		USN-2	As a user, I will receive confirmation email once I have registered for the application	6	High	Team Lead Member 1 Member 2 Member 3
Sprint-4		USN-3	As a user, I can register for the application using my Gmail account.	6	Low	Team Lead Member 1 Member 2 Member 3
Sprint-1	Login	USN-4	As a user, I can register for the application by entering my credentials.	5	Medium	Team Lead Member 1 Member 2 Member 3
Sprint-3	Dashboard	USN-5	As a user, I can view my past activities in the application.	6	High	Team Lead Member 1 Member 2 Member 3
Sprint-2	Entry form	USN-6	As a user, I need to enter my pre-diagnostic test results	7	High	Team Lead Member 1 Member 2

						Member 3
Sprint-3	Report	USN-7	As a user, I can view the report generated by the application.	7	High	Team Lead Member 1 Member 2 Member 3
Sprint-3	Remedies	USN-8	As a user, I will receive the treatment for my symptoms	6	Medium	Team Lead Member 1 Member 2 Member 3
Sprint-4	Queries	USN-9	As a customer care executive, I should be able to assist the users who are facing any sort of difficulties.	5	Low	Team Lead Member 1 Member 2 Member 3
Sprint-4	Feedback	USN-10	As a customer care executive, I should get input for the tools enhancement.	7	Low	Team Lead Member 1 Member 2 Member 3
Sprint-2	Feature importance	USN-11	As an administrator, I should identify the most significant factors that lead to CKD based on the present thread	6	High	Team Lead Member 1 Member 2 Member 3
Sprint-2	Train model	USN-12	As an administrator, I should use the most suitable ML model for the detection of CKD.	5	High	Team Lead Member 1 Member 2 Member 3

6.2 Sprint delivery schedule

A milestone list is a document used in project management that lists each project milestone. A project milestone is a noteworthy occasion or turning point. Since it just represents a single instant in time, milestones should be given a duration of 0 when scheduling.

A milestone is a particular point in a project's life cycle that is used to gauge progress made toward the end result. A project's start and finish dates, external reviews and input, budget checks, the submission of a major deliverable, etc. are all indicated by milestones in project management.

A task is an activity that needs to be finished in order to reach a milestone, whereas a milestone is a point on the schedule that has one clearly defined deliverable. A task or activity has a beginning and ending date. While a milestone is a single date on which a deliverable is finished, it is two dimensional.

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	19	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	19	6 Days	31 Oct 2022	05 Nov 2022	20	5 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	19	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022

So for our team the velocity is:

$$\text{Sprint 1 AV} = \text{Sprint duration} / \text{Velocity} = 6 / 19 = .315$$

$$\text{Sprint 2 AV} = \text{Sprint duration} / \text{Velocity} = 6 / 19 = .315$$

$$\text{Sprint 3 AV} = \text{Sprint duration} / \text{Velocity} = 6 / 20 = .3$$

$$\text{Sprint 4 AV} = \text{Sprint duration} / \text{Velocity} = 6 / 19 = .315$$

6.3 Reports from JIRA

 **CKD predictor**
Software project

PLANNING

 Roadmap

 **Board**

> DEVELOPMENT

 Project pages

 Add shortcut

 Project settings

You're in a team-managed project

[Learn more](#)

Projects / CKD predictor

CP

TO DO 7 ISSUES

USN-3 Register using Gmail account

☒ CP-3

USN-5 View Dashboard

☒ CP-5

USN-7 View diagnosis report

☒ CP-7

USN-8 Remedies for diagnosis

☒ CP-8

USN-9 Customer Q&A

☒ CP-9

USN-10 Customer feedback

☒ CP-10

USN-12 Select optimal model

☒ CP-12

IN PROGRESS 3 ISSUES

USN-11 Feature selection for model

☒ CP-11

USN-2 Confirmation Email on registration

☒ CP-2

USN-6 Entry form for pre-diagnostic test

☒ CP-6

DONE 2 ISSUES 

USN-1 Registration Page

☒ CP-1 

USN-4 Login page for registered users

☒ CP-4 

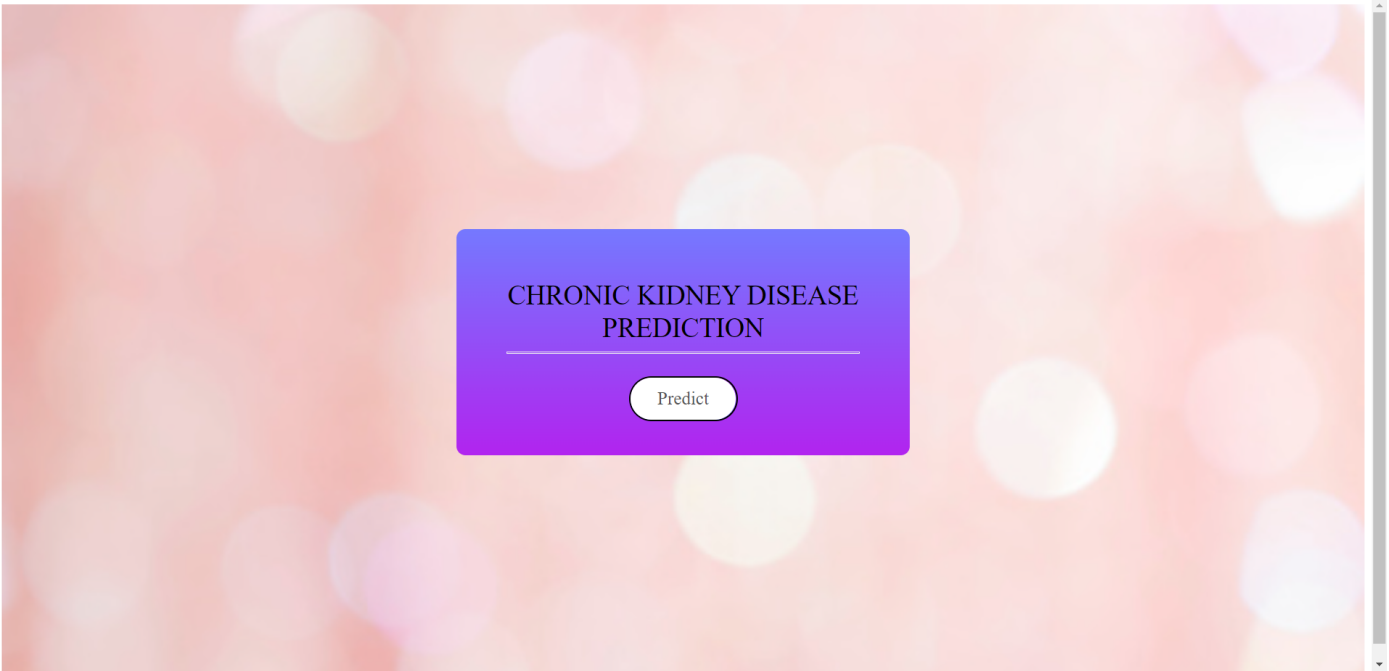


7. Coding and Solution

7.1 Home

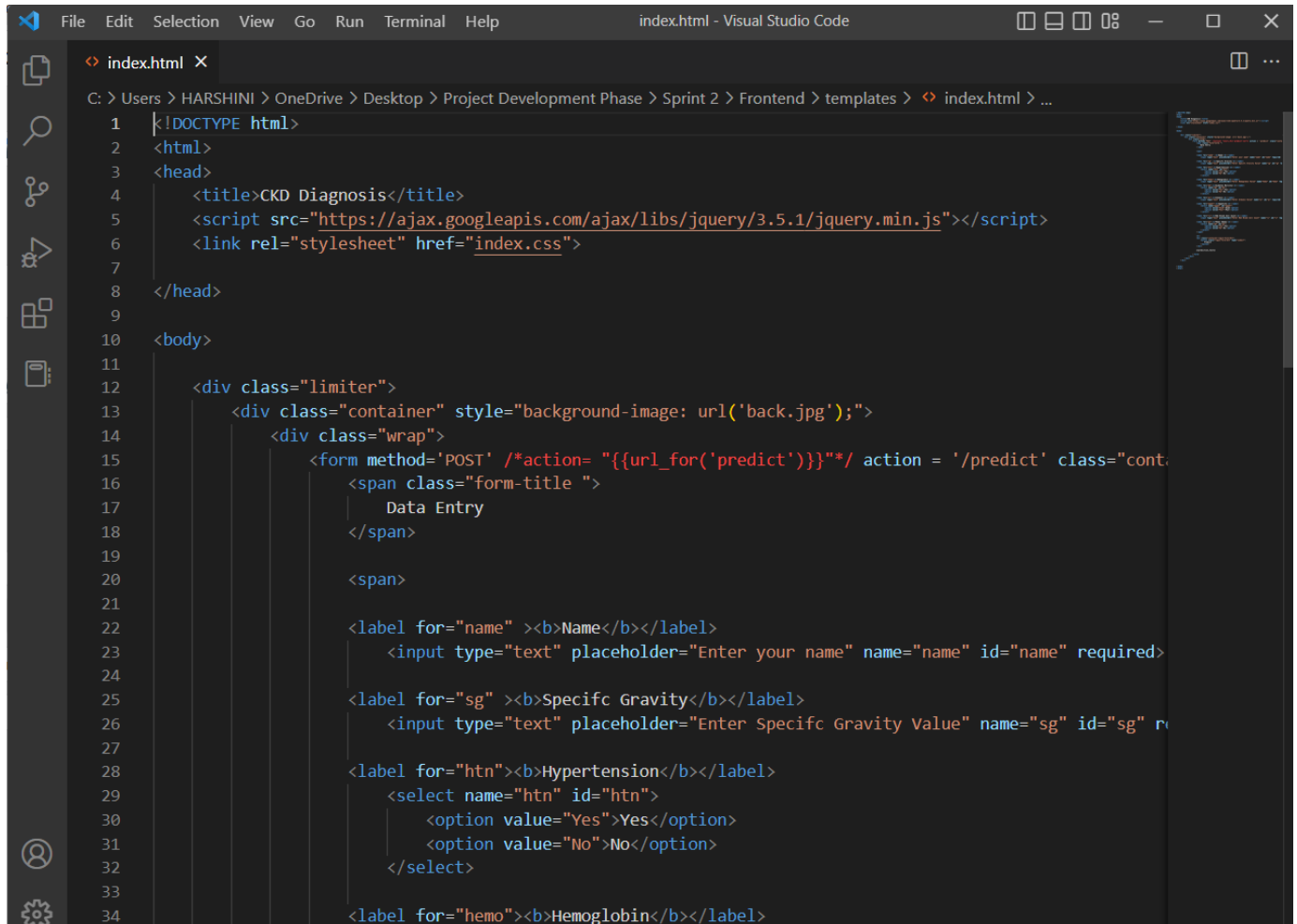
Home.html

```
home.html X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > templates > home.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Home</title>
5      <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
6      <link rel="stylesheet" href="index.css">
7
8  </head>
9  <body>
10     <div class="limiter">
11         <div class="container" style="background-image: url('back.jpg');">
12             <div class="wrap">
13                 <span class="form-title ">
14                     CHRONIC KIDNEY DISEASE PREDICTION
15                 </span>
16                 <hr>
17                 <span>
18                     <div class="container-login-form-btn">
19                         <button class="login-form-btn" onclick="document.location='index.html'">
20                             Predict
21                         </button>
22                     </div>
23                 </span>
24             </div>
25         </div>
26     </div>
27 </body>
28 </html>
29
```



7.2 Index and Report

Index.html



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>CKD Diagnosis</title>
5   <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
6   <link rel="stylesheet" href="index.css">
7 </head>
8
9
10 <body>
11
12   <div class="limiter">
13     <div class="container" style="background-image: url('back.jpg');">
14       <div class="wrap">
15         <form method='POST' /*action= "{{url_for('predict')}}"*/ action = '/predict' class="cont
16         <span class="form-title">
17           Data Entry
18         </span>
19
20         <span>
21
22           <label for="name"><b>Name</b></label>
23           <input type="text" placeholder="Enter your name" name="name" id="name" required>
24
25           <label for="sg"><b>Specifc Gravity</b></label>
26           <input type="text" placeholder="Enter Specifc Gravity Value" name="sg" id="sg" r
27
28           <label for="htn"><b>Hypertension</b></label>
29           <select name="htn" id="htn">
30             <option value="Yes">Yes</option>
31             <option value="No">No</option>
32           </select>
33
34           <label for="hemo"><b>Hemoglobin</b></label>
```

```

index.html X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > templates > index.html > ...
34 <label for="hemo"><b>Hemoglobin</b></label>
35 <input type="text" placeholder="Enter Hemoglobin Value" name="hemo" id="hemo" required>
36
37 <label for="dm"><b>Diabetes Mellitus</b></label>
38 <select name="dm" id="dm">
39 | <option value="Yes">Yes</option>
40 | <option value="No">No</option>
41 </select>
42
43 <label for="al"><b>Albumin</b></label>
44 <input type="text" placeholder="Enter Albumin Value" name="al" id="al" required>
45
46 <label for="appet"><b>Appetite</b></label>
47 <select name="appet" id="appet">
48 | <option value="Good">Good</option>
49 | <option value="Poor">Poor</option>
50 </select>
51
52 <label for="rc"><b>Red Blood Cell Count</b></label>
53 <input type="text" placeholder="Enter Red Blood Cell Count" name="rc" id="rc" required>
54
55 <label for="pe"><b>Pedal Edema</b></label>
56 <select name="pe" id="pe">
57 | <option value="Yes">Yes</option>
58 | <option value="No">No</option>
59 </select>
60 </span>
61
62 <hr>
63 <div class="container-login-form-btn">
64 | <button class="login-form-btn" type="submit">
65 | | Predict
66 | </button>
67 </div>

```

```

index.html X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > templates >
53
54
55         <label for="pe"><b>Pedal Edema</b></label>
56         <select name="pe" id="pe">
57             <option value="Yes">Yes</option>
58             <option value="No">No</option>
59         </select>
60     </span>
61
62     <hr>
63     <div class="container-login-form-btn">
64         <button class="login-form-btn" type="submit">
65             Predict
66         </button>
67     </div>
68
69     </form>
70 </div>
71 </div>
72 </div>
73
74
75 </body>
76 </html>
77

```

DATA ENTRY

Name

Specific Gravity

Hypertension

Hemoglobin

Diabetes Mellitus

Albumin

Appetite
Good

Red Blood Cell Count
Enter Red Blood Cell Count

Pedal Edema
Yes

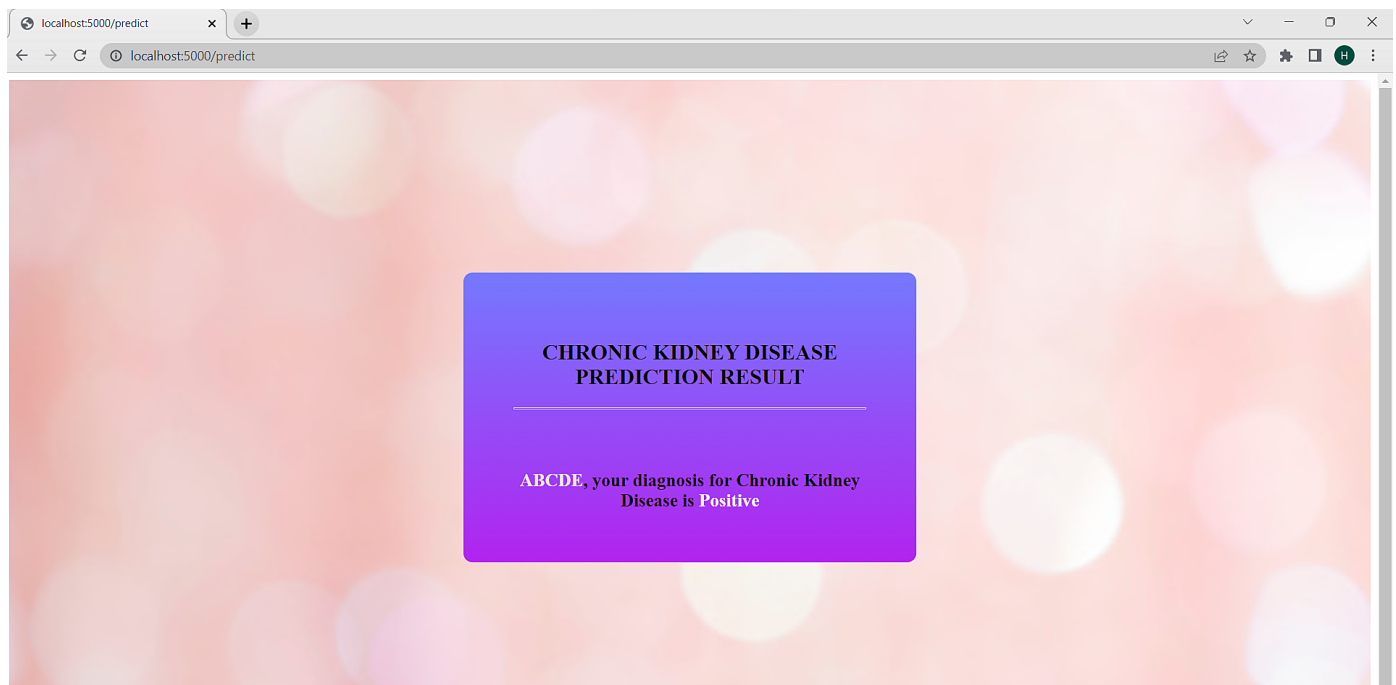
Predict

Report.html

```

1  <!DOCTYPE html>
2  <html>
3      <head>
4          <link rel="stylesheet" href="index.css">
5      </head>
6      <body>
7          <div class = "limiter">
8              <div class="container" style="background-image: url('back.jpg');">
9                  <div class="wrap">
10                     <h2 align="center">CHRONIC KIDNEY DISEASE PREDICTION RESULT</h2>
11                     <hr>
12                     <div style="font-weight: bold; font-size:20px; color:black;">
13                         <br>
14                         <p align="center"><span style="font-weight: bold; color:rgb(237, 228, 228);">{
15                         <span style="font-weight: bold; color:rgb(245, 238, 238);">{{ prediction_text
16                         </div>
17                     </table>
18                 </div>
19             </div>
20         </div>
21     </body>
22 </html>

```



7.3 Flask Integration and Deployment

Integrating Flask with Pickle

```
app-pkl.py X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > app-pkl.py >
1  from flask import Flask,render_template,request, jsonify
2  from flask_cors import CORS
3  import pickle
4  import numpy as np
5
6
7  model=pickle.load(open('kidney.pkl','rb'))
8  app=Flask(__name__, static_url_path='')
9  CORS(app)
10
11 @app.route('/', methods=['GET'])
12 def index():
13     return render_template('home.html')
14
15 @app.route('/predict',methods=['POST'])
16 def predict():
17     name = request.form['name']
18
19     v1 = float(request.form['sg'])
20
21     v2 = request.form['htn']
22     if(v2 == 'No'):
23         v2 = 0
24     if v2 == 'Yes':
25         v2 = 1
26
27     v3 = float(request.form['hemo'])
28
```

```

app.pkl.py X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > app.pkl.py >
26
27     v3 = float(request.form['hemo'])
28
29     v4 = request.form['dm']
30     if(v4 == 'No'):
31         v4 = 0
32     if v4 == 'Yes':
33         v4 = 1
34
35     v5 = int(request.form['al'])
36
37     v6 = request.form['appet']
38     if(v6 == 'Poor'):
39         v6 = 0
40     if v6 == 'Good':
41         v6 = 1
42
43     v7 = float(request.form['rc'])
44
45     v8 = request.form['pe']
46     if(v8 == 'No'):
47         v8 = 0
48     if v8 == 'Yes':
49         v8 = 1
50
51     result = model.predict(np.array([v1,v2,v3,v4,v5,v6,v7,v8]).reshape(1,8))
52     d = {0:"Negative",1:"Positive"}
53     output = d[result[0]]
54     return render_template('report.html',prediction_text = output, your_name = name)
55
56 if __name__ == '__main__':
57     app.run(debug = True)
58

```

Pickle

For our project we are choosing Random Forest Classifier.

```

import pickle
pickle.dump(RandomForest, open('kidney.pkl', 'wb'))

```


Integrating Flask with IBM Watson Machine Learning Studio

```
app.py
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > app.py > ...
1  from flask import Flask,render_template,request, jsonify
2  from flask_cors import CORS
3  import pickle
4  import numpy as np
5  import requests
6  import json
7
8  # NOTE: you must manually set API_KEY below using information retrieved from your IBM Cloud account.
9  API_KEY = "53Wbj0Dkinpv78FeGLOzcscvYlnmhVF4M4vgU5h1rV_1"
10 token_response = requests.post('https://iam.cloud.ibm.com/identity/token', data={"apikey":
11 | API_KEY, "grant_type": 'urn:ibm:params:oauth:grant-type:apikey'})
12 mltoken = token_response.json()["access_token"]
13
14 header = {'Content-Type': 'application/json', 'Authorization': 'Bearer ' + mltoken}
15
16 app=Flask(__name__, static_url_path='')
17 CORS(app)
18
19 @app.route('/', methods=['GET'])
20 def index():
21     return render_template('home.html')
22
23 @app.route('/predict',methods=['POST'])
24 def predict():
25     name = request.form['name']
26
27     v1 = float(request.form['sg'])
28
29     v2 = request.form['htn']
30     if(v2 == 'No'):
31         v2 = 0
32     if v2 == 'Yes':
33         v2 = 1
```

```
app.py X
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > app.py > ...

36     v3 = float(request.form['hemo'])
37
38     v4 = request.form['dm']
39     if(v4 == 'No'):
40         v4 = 0
41     if v4 == 'Yes':
42         v4 = 1
43
44     v5 = int(request.form['al'])
45
46     v6 = request.form['appet']
47     if(v6 == 'Poor'):
48         v6 = 0
49     if v6 == 'Good':
50         v6 = 1
51
52     v7 = float(request.form['rc'])
53
54     v8 = request.form['pe']
55     if(v8 == 'No'):
56         v8 = 0
57     if v8 == 'Yes':
58         v8 = 1
59
60     t = [[v1, int(v2), v3, int(v4), v5, int(v6), v7, int(v8)]]
61     print(t)
62     payload_scoring = {"input_data": [{"field": ["sg", "htn", "hemo", "dm", "al", "appet", "rc", "pe"], "values": t}]}
```

```
app.py
C: > Users > HARSHINI > OneDrive > Desktop > Project Development Phase > Sprint 2 > Frontend > app.py > ...

50
51     v7 = float(request.form['rc'])
52
53     v8 = request.form['pe']
54     if(v8 == 'No'):
55         v8 = 0
56     if v8 == 'Yes':
57         v8 = 1
58
59     t = [[v1, int(v2), v3, int(v4), v5, int(v6), v7, int(v8)]]
60     print(t)
61     payload_scoring = {"input_data": [{"field": ["sg", "htn", "hemo", "dm", "al", "appet", "rc", "pe"], "values": t}]}
62
63     response_scoring = requests.post('https://us-south.ml.cloud.ibm.com/ml/v4/deployments/2482ca43-5c91-4ab8-8959-ec30a7e635f7/predictions?version=2022-11-17',
64                                     , json=payload_scoring, headers={'Authorization': 'Bearer ' + mltoken})
65
66     predictions = response_scoring.json()
67     print(predictions)
68
69     pred = predictions["predictions"][0]["values"][0][0]
70     if pred == 0:
71         output = "Negative"
72     else:
73         output = "Positive"
74     print(output)
75     return render_template('report.html', prediction_text = output, your_name = name)
76
77 if __name__ == '__main__':
78     app.run(debug = True)
79
```

Deployed Jupyter Notebook in IBM

IBM Watson Studio

Search in your workspaces

Buy

Harini P's Account

Projects / Custom Model Deployment / Model_Selection

```
In [3]: # Importing Libraries:
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

In [4]: # for displaying all feature from dataset:
pd.pandas.set_option('display.max_columns', None)
```

```
In [7]: import os, types
import pandas as pd
from botocore.client import Config
import ibm_boto3

def __iter__(self): return 0

# @hidden_cell
# The following code accesses a file in your IBM Cloud Object Storage. It includes your credentials.
# You might want to remove those credentials before you share the notebook.
cos_client = ibm_boto3.client(service_name='s3',
                              ibm_api_key_id='bsqC-Uh6ZlzxwZxQKBw7znj4h0016_HngFfMdK5tR5jQ',
                              ibm_auth_endpoint="https://iam.cloud.ibm.com/oidc/token",
                              config=Config(signature_version='oauth')),
                              endpoint_url='https://s3.private.us.cloud-object-storage.appdomain.cloud')

bucket = 'custommodeldeployment-donotdelete-pr-c8epradx27d5vu'
object_key = '4.chronickidneydisease.csv'

body = cos_client.get_object(Bucket=bucket,Key=object_key)['Body']
# add missing __iter__ method, so pandas accepts body as file-like object
if not hasattr(body, "__iter__"): body.__iter__ = types.MethodType( __iter__, body )

dataset = pd.read_csv(body)
dataset.head()
```

	id	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr	bu	sc	sod	pot	hemo	pcv	wc	rc	htn	dm	cad	appet	pe	ane	classification
0	0	48.0	80.0	1.020	1.0	0.0	NaN	normal	notpresent	notpresent	121.0	36.0	1.2	NaN	NaN	15.4	44	7800	5.2	yes	yes	no	good	no	no	ckd
1	1	7.0	50.0	1.020	4.0	0.0	NaN	normal	notpresent	notpresent	NaN	18.0	0.8	NaN	NaN	11.3	38	6000	NaN	no	no	no	good	no	no	ckd
2	2	62.0	80.0	1.010	2.0	3.0	normal	normal	notpresent	notpresent	423.0	53.0	1.8	NaN	NaN	9.6	31	7500	NaN	no	yes	no	poor	no	yes	ckd
3	3	48.0	70.0	1.005	4.0	0.0	normal	abnormal	present	notpresent	117.0	56.0	3.8	111.0	2.5	11.2	32	6700	3.9	yes	no	no	poor	yes	yes	ckd
4	4	51.0	80.0	1.010	2.0	0.0	normal	normal	notpresent	notpresent	106.0	26.0	1.4	NaN	NaN	11.6	35	7300	4.6	no	no	no	good	no	no	ckd

```
In [34]: !pip install ibm_watson_machine_learning

Requirement already satisfied: ibm_watson_machine_learning in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (1.0.257)
Requirement already satisfied: requests in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (2.26.0)
Requirement already satisfied: lomond in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (0.3.3)
Requirement already satisfied: certifi in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (2022.9.24)
Requirement already satisfied: pandas<1.5.0,>=0.24.2 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (1.3.4)
Requirement already satisfied: packaging in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (21.3)
Requirement already satisfied: tabulate in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (0.8.9)
Requirement already satisfied: urllib3 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (1.26.7)
Requirement already satisfied: importlib-metadata in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (4.8.2)
Requirement already satisfied: ibm-cos-sdk==2.11.* in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm_watson_machine_learning) (2.11.0)
Requirement already satisfied: ibm-cos-sdk-core==2.11.0 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm-cos-sdk==2.11.*->ibm_watson_machine_learning) (2.11.0)
Requirement already satisfied: ibm-cos-sdk-s3transfer==2.11.0 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm-cos-sdk==2.11.*->ibm_watson_machine_learning) (2.11.0)
Requirement already satisfied: jmespath<1.0.0,>=0.7.1 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm-cos-sdk==2.11.*->ibm_watson_machine_learning) (0.10.0)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from ibm-cos-sdk-core==2.11.0->ibm-cos-sdk==2.11.*->ibm_watson_machine_learning) (2.8.2)
Requirement already satisfied: pytz>=2017.3 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from pandas<1.5.0,>=0.24.2->ibm_watson_machine_learning) (2021.3)
Requirement already satisfied: numpy>=1.17.3 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from pandas<1.5.0,>=0.24.2->ibm_watson_machine_learning) (1.20.3)
Requirement already satisfied: six>=1.5 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from python-dateutil<3.0.0,>=2.1->ibm-cos-sdk-core==2.11.0->ibm-cos-sdk==2.11.*->ibm_watson_machine_learning) (1.15.0)
Requirement already satisfied: charset-normalizer==2.0.0 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from requests->ibm_watson_machine_learning) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from requests->ibm_watson_machine_learning) (3.3)
Requirement already satisfied: zipp>=0.5 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from importlib-metadata->ibm_watson_machine_learning) (3.6.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /opt/conda/envs/Python-3.9/lib/python3.9/site-packages (from packaging->ibm_watson_machine_learning) (3.0.4)
```

```

In [35]: #Connecting machine learning service as API client
from ibm_watson_machine_learning import APIClient
wml_credentials = {
    "url" : "https://us-south.ml.cloud.ibm.com",
    "apikey" : "53WbjoDkinpv78FeGLOzcscvYlnmhVF4M4vgU5h1rV_1"
}
client = APIClient(wml_credentials)

In [36]: #Creating deployment space
def guid_from_space_name(client, space_name):
    space = client.spaces.get_details()
    return(next(item for item in space["resources"] if item["entity"]["name"] == space_name)["metadata"]["id"])

In [39]: space_uid = guid_from_space_name(client, "models")
print("Space UID = " + space_uid)

Space UID = 940e0787-3a5c-4038-83f9-37e5f42ea279

In [40]: client.set.default_space(space_uid)

Out[40]: 'SUCCESS'

```

In [41]: `client.software_specifications.list()`

```

-----
NAME                               ASSET_ID                               TYPE
default_py3.6                      0062b8c9-8b7d-44a0-a9b9-46c416adcbd9 base
kernel-spark3.2-scala2.12          020d69ce-7ac1-5e68-ac1a-31189867356a base
pytorch-onnx_1.3-py3.7-edt        069ea134-3346-5748-b513-49120e15d288 base
scikit-learn_0.20-py3.6           09c5a1d0-9c1e-4473-a344-eb7b665ff687 base
spark-mllib_3.0-scala_2.12         09f4cff0-90a7-5899-b9ed-1ef348aebdee base
pytorch-onnx_rt22.1-py3.9          0b848dd4-e681-5599-be41-b5f6fccc6471 base
ai-function_0.1-py3.6              0cdb0f1e-5376-4f4d-92dd-da3b69aa9bda base
shiny-r3.6                         0e6e79df-875e-4f24-8ae9-62dcc2148306 base
tensorflow_2.4-py3.7-horovod       1092590a-307d-563d-9b62-4eb7d64b3f22 base
pytorch_1.1-py3.6                  10ac12d6-6b30-4ccd-8392-3e922c096a92 base
tensorflow_1.15-py3.6-ddl          111e41b3-de2d-5422-a4d6-bf776828c4b7 base
autoai-kb_rt22.2-py3.10            125b6d9a-5b1f-5e8d-972a-b251688ccf40 base
runtime-22.1-py3.9                 12b83a17-24d8-5082-900f-0ab31fbfd3cb base
scikit-learn_0.22-py3.6            154010fa-5b3b-4ac1-82af-4d5ee5abbc85 base
default_r3.6                       1b70aec3-ab34-4b87-8aa0-a4a3c8296a36 base
pytorch-onnx_1.3-py3.6             1bc6029a-cc97-56da-b8e0-39c3880dbbe7 base
kernel-spark3.3-r3.6               1c9e5454-f216-59dd-a20e-474a5cdf5988 base
pytorch-onnx_rt22.1-py3.9-edt      1d362186-7ad5-5b59-8b6c-9d0880bde37f base
tensorflow_2.1-py3.6               1eb25b84-d6ed-5dde-b6a5-3fbdf1665666 base
spark-mllib_3.2                    20047f72-0a98-58c7-9ff5-a77b012eb8f5 base
tensorflow_2.4-py3.8-horovod       217c16f6-178f-56bf-824a-b19f20564c49 base
runtime-22.1-py3.9-cuda            26215f05-08c3-5a41-a1b0-da66306ce658 base
do_py3.8                           295addb5-9ef9-547e-9bf4-92ae3563e720 base
autoai-ts_3.8-py3.8                2aa0c932-798f-5ae9-abd6-15e0c2402fb5 base
tensorflow_1.15-py3.6              2b73a275-7cbf-420b-a912-eae7f436e0bc base
kernel-spark3.3-py3.9              2b7961e2-e3b1-5a8c-a491-482c8368839a base
pytorch_1.2-py3.6                  2c8ef57d-2687-4b7d-acce-01f94976dac1 base
spark-mllib_2.3                     2e51f700-bca0-4b0d-88dc-5c6791338875 base
pytorch-onnx_1.1-py3.6-edt         32983cea-3f32-4400-8965-dde874a8d67e base
spark-mllib_3.0-py37               36507ebe-8770-55ba-ab2a-eafe787600e9 base
spark-mllib_2.4                     390d21f8-e58b-4fac-9c55-d7ceda621326 base
autoai-ts_rt22.2-py3.10            396b2e83-0953-5b86-9a55-7ce1628a406f base
xgboost_0.82-py3.6                 39e31acd-5f30-41dc-ae44-60233c80306e base
pytorch-onnx_1.2-py3.6-edt         40589d0a-7019-4a78-8daa-fb03b6f1fa12 base

```

xgboost_0.90-py3.6	4ff8d6c2-1343-4c18-85e1-689c965304d3	base
pytorch-onnx_1.1-py3.6	50f95b2a-bc16-43bb-bc94-b0bed208c60b	base
autoai-ts_3.9-py3.8	52c57136-80fa-572e-8728-a5e7cbb42cde	base
spark-mllib_2.4-scala_2.11	55a70f99-7320-4be5-9fb9-9edb5a443af5	base
spark-mllib_3.0	5c1b0ca2-4977-5c2e-9439-ffd44ea8ffe9	base
autoai-obm_2.0	5c2e37fa-80b8-5e77-840f-d912469614ee	base
spss-modeler_18.1	5c3cad7e-507f-4b2a-a9a3-ab53a21dee8b	base
cuda-py3.8	5d3232bf-c86b-5df4-a2cd-7bb870a1cd4e	base
runtime-22.2-py3.10-xc	5e8cddff-db4a-5a6a-b8aa-2d4af9864dab	base
autoai-kb_3.1-py3.7	632d4b22-10aa-5180-88f0-f52dfb6444d7	base

Note: Only first 50 records were displayed. To display more use 'limit' parameter.

```
In [50]: software_spec_uid = client.software_specifications.get_uid_by_name("runtime-22.1-py3.9")
software_spec_uid
```

```
Out[50]: '12b83a17-24d8-5082-900f-0ab31fbfd3cb'
```

```
In [53]: #Store model
model_details = client.repository.store_model(model = RandomForest, meta_props = {
    client.repository.ModelMetaNames.NAME: "CKD_Model",
    client.repository.ModelMetaNames.TYPE: 'scikit-learn_1.0',
    client.repository.ModelMetaNames.SOFTWARE_SPEC_UID: software_spec_uid
})
model_id = client.repository.get_model_id(model_details)
```

```
In [54]: model_id
```

```
Out[54]: 'f02f51fa-6cd5-45b4-9541-2664bb8db945'
```

```
1 [35]: model_id
```

```
Out[35]: 'f320aad5-519b-4c8c-868c-30507c518b15'
```

```
1 [39]: model_details
```

```
Out[39]: {'entity': {'hybrid_pipeline_software_specs': [],
  'software_spec': {'id': '12b83a17-24d8-5082-900f-0ab31fbfd3cb',
    'name': 'runtime-22.1-py3.9'},
  'type': 'scikit-learn_1.0'},
  'metadata': {'created_at': '2022-11-18T16:01:14.360Z',
    'id': 'f320aad5-519b-4c8c-868c-30507c518b15',
    'modified_at': '2022-11-18T16:01:17.444Z',
    'name': 'CKD_Model',
    'owner': 'IBMid-668000FWR5',
    'resource_key': '355461e6-197f-4a1a-a2c2-b79ce8ca9888',
    'space_id': '940e0787-3a5c-4038-83f9-37e5f42ea279'},
  'system': {'warnings': []}}
```

```
1 [38]: # Set meta
deployment_props = {
    client.deployments.ConfigurationMetaNames.NAME: "CKD_Model",
    client.deployments.ConfigurationMetaNames.ONLINE: {}
}
```

```
In [38]: # Set meta
deployment_props = {
    client.deployments.ConfigurationMetaNames.NAME: "CKD_Model",
    client.deployments.ConfigurationMetaNames.ONLINE: {}
}
```

```
In [40]: # Deploy
deployment = client.deployments.create(
    artifact_uid=model_id,
    meta_props=deployment_props
)
```

```
#####

Synchronous deployment creation for uid: 'f320aad5-519b-4c8c-868c-30507c518b15' started

#####

initializing
Note: online_url is deprecated and will be removed in a future release. Use serving_urls instead.
.
ready

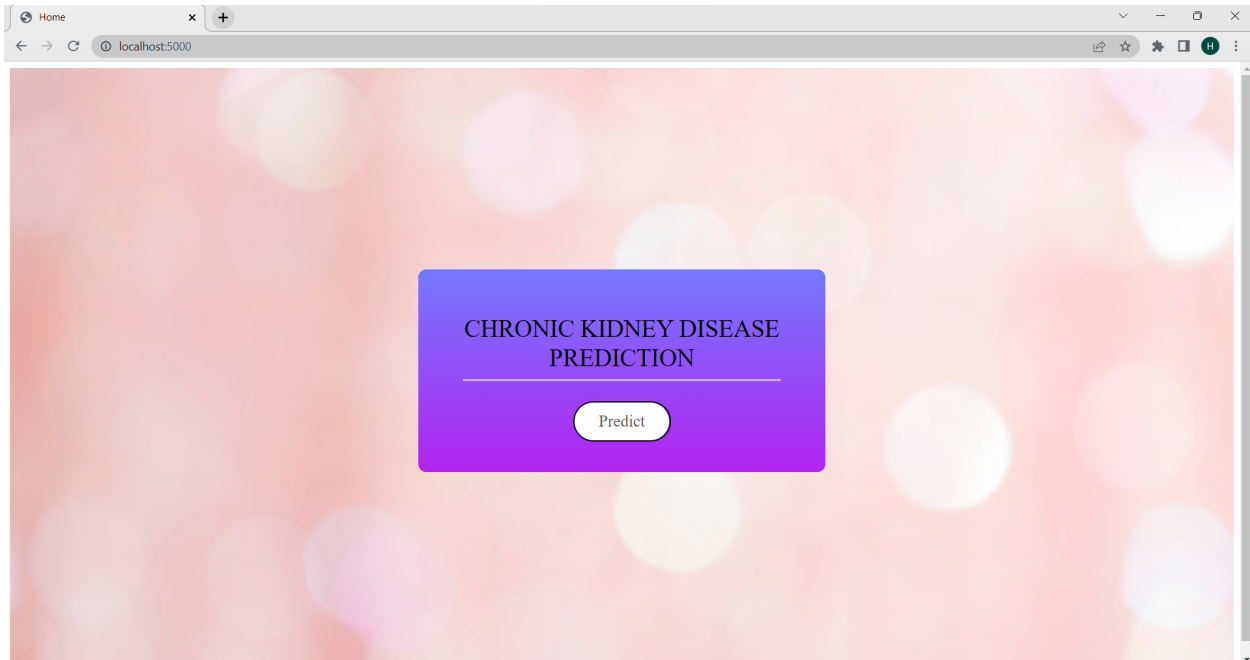
-----
Successfully finished deployment creation, deployment_uid='3d956463-34fa-4458-8761-36dac17572a8'
-----
```

8. Testing

8.1 Test Cases

Test case ID	Feature Type	Test Scenario	Pre-Requisite	Steps To Execute	Test Data	Expected Result	Actual Result	Status	TC for Automation(Y/N)
Home_TC_OO1	Functional	Verify user can access the Home page	home.html	1.Enter the home website. 2.See if website is seen properly.	localhost:5000	Home page is available.	Working as expected	Pass	N
Home_TC_OO2	Functional	Verify user can click the Predict button	home.html	1.Click the Predict button	localhost:5000	Page gets redirected to index.html page.	Working as expected	Pass	N
Index_TC_OO1	UI	Verify the UI elements	index.html	1.In the index.html page, enter the details asked for prediction. 2.Required fields must be filled.	localhost:5000/index.html	User enters relevant input.	Working as expected	Pass	N
Index_TC_OO2	Functional	Verify user data is given as input to prediction model	1.report.html 2.app.py	User will enter their details asked.	localhost:5000/index.html	Application sends data to app.py	Working as expected	Pass	N
Report_TC_OO1	Functional	Verify user is able to see their results	1.report.html 2.app.py	1.User will be able to view the prediction result.	localhost:5000/report.html	Application shows the result that is predicted	Working as expected	Pass	N

8.2 Sample Tests



CKD Diagnosis x +

localhost:5000/index.html

Home

DATA ENTRY

Name

ABCDE

Specific Gravity

1.02

Hypertension

Yes

Hemoglobin

15.4

Diabetes Mellitus

Yes

Albumin

1

Appetite

Good

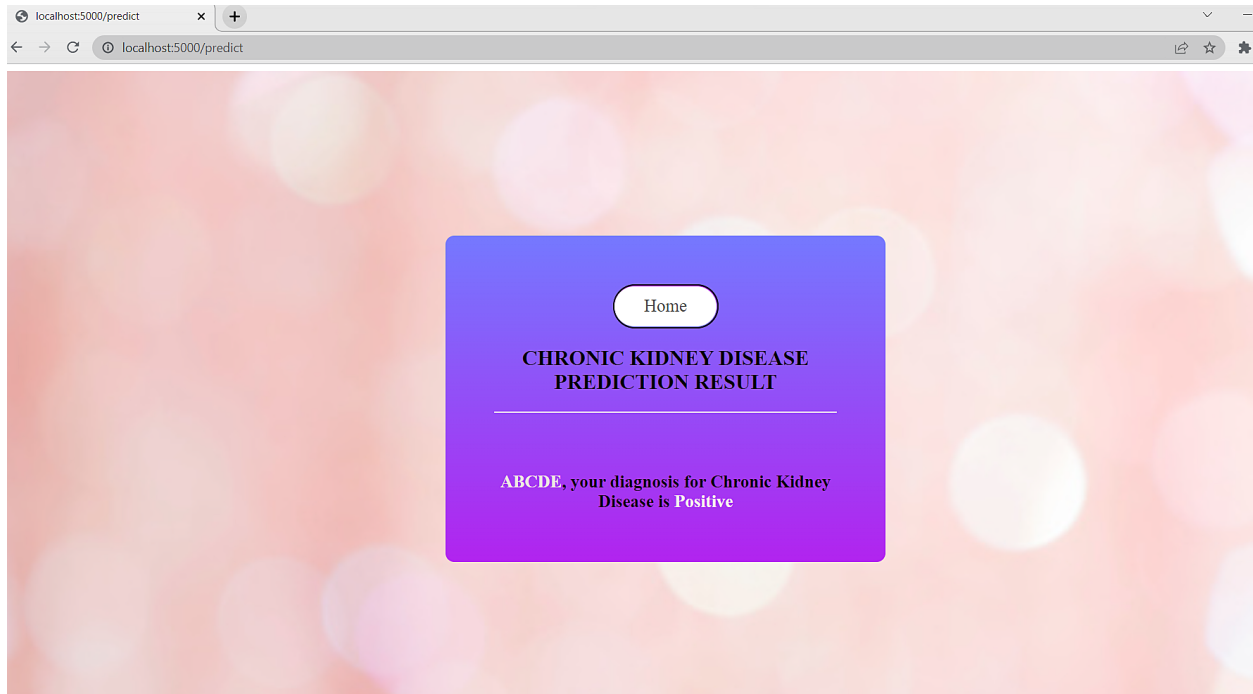
Red Blood Cell Count

5.2

Pedal Edema

No

Predict



8.3 User Acceptance Testing (UAT)

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the [Early Detection of Chronic Kidney Disease using Machine Learning] project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	2	3	1	1	7
Duplicate	3	2	1	0	6
External	2	1	0	1	4
Fixed	1	2	1	1	5
Not Reproduced	0	0	0	0	0
Skipped	0	0	0	0	0
Won't Fix	0	0	0	0	0
Totals	8	8	3	3	22

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Home Page	1	0	0	1
User Details	2	0	0	2
CKD Prediction	1	0	0	1
No CKD Prediction	1	0	0	1
Final Report Output	1	0	0	1

9.Results

9.1 Performance metrics (Random Forest Classifier)

```
# Random Forest Classifier
from sklearn.ensemble import RandomForestClassifier
RandomForest = RandomForestClassifier()
RandomForest = RandomForest.fit(X_train,y_train)

# Predict
y_pred = RandomForest.predict(X_test)

rf = accuracy_score(y_test,y_pred)
print('Accuracy:', rf*100)

print(confusion_matrix(y_test,y_pred))
print(classification_report(y_test,y_pred))
```

Accuracy: 100.0
[[58 0]
[0 62]]

		precision	recall	f1-score	support
	0	1.00	1.00	1.00	58
	1	1.00	1.00	1.00	62
	accuracy			1.00	120
	macro avg	1.00	1.00	1.00	120
	weighted avg	1.00	1.00	1.00	120

Performance of the Random Forest Classifier model is evaluated through the accuracy score, confusion matrix and classification report. The values of precision, recall, f1 - score and support are used to analyze the performance of the model.

10. Advantages and disadvantages

Advantages:

A major health issue because of its rising incidence is chronic kidney disease (CKD). The syndrome is also known as chronic renal disease, which is characterized by a progressive decline in kidney function over time.

It would be far more beneficial to employ a machine learning algorithm to develop a better testing procedure that might possibly detect CKD in its early stages.

Detecting chronic kidney disease at an early stage also has the following advantages:

1. Greater cost savings in hospital testing
2. Increased percentage of recovery
3. Lesser complications

Disadvantages:

Chronic kidney detection comes with a lot of advantages but there are some disadvantages that are present in this model of prediction. The disadvantages that are present in the chronic kidney disease prediction model are as follows:

1. The prediction may be wrong, which may lead to further implications.
2. The prediction model is a web application so it needs to be maintained and needs to be tested regularly to see if there are any bugs present in the application.

11. Conclusion

As the name says, chronic kidney disease is a chronic condition. Any chronic condition would make the individual miserable and endure longer until they lost their livelihood. It is extremely reckless and puts a person's life in danger if in these situations the disease goes undiscovered in the early stages, when it can be treated by medical institutions.

Finding the best answer is crucial in these situations, which is why a machine learning model is used for early diagnosis and prediction of chronic kidney disease. If the disease is discovered in its early stages, it can be immediately treated, drastically reducing the risk of contracting it. Consider the traditional method of diagnosing kidney disease, which involves a blood test. Blood test results take longer than anticipated, but blood tests are only one stage in the process; there are many more tests that must be done, all of which can take time.

When this occurs, the model prediction helps the medical team treat the patient if he or she is susceptible to the disease sooner and more effectively.

Thus, in order to diagnose the patient quickly and begin the appropriate treatment to cure them if detected, early identification of chronic kidney disease is crucial in today's hospitals.

12. Future scope

The project can be further developed in the future to not only predict chronic kidney disease but various other chronic diseases by changing the dataset and the user inputs.

The model can be improved by training and testing it with more diverse dataset, in order to get more accurate prediction.

13. Appendix

Github link: <https://github.com/IBM-EPBL/IBM-Project-25054-1659953169>

Source code: <https://github.com/IBM-EPBL/IBM-Project-25054-1659953169/tree/main/Final%20deliverables/templates>